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Manual / Product Information standard mFRR product (MARI-product) for BSPs

Requirements and procedures for standard mFRR product (MARI-product) delivery

Foreword and reader's guide

For maintaining the real-time power balance of the Netherlands, TenneT¹ uses Frequency Restoration Reserves (FRR). A distinction can be made between automatic Frequency Restoration Reserve (aFRR) and mFRRda (manual Frequency Restoration Reserve directly activated). In case of incidents and substantial long-lasting power imbalances TenneT manually activates *mFRRda*. This document refers to the *mFRR standard product*, also referred to as *MARI-product* in the Netcode². These products are offered to TenneT by market parties in their role as Balancing Service Provider (BSP).

The mFRR standard product (MARI-product) is not intended for use by TenneT to solve imbalances in the Dutch control area in normal system state. It is envisioned to make use of it in non-normal system state, but such procedure is not yet defined and therefore does not yet apply to this document.

Therefore, for the scope of validity of this document, mFRR standard product activations are the result of the MARI optimization in response to foreign TSOs' demand and available cross-zonal capacity (CZC), independently on the state of Dutch electricity system.

This document describes both the operational process related to the provision and activation of mFRR standard product, as well as the requirements the BSP must meet.

¹ Because this document discusses the *Dutch* balancing situation, "TenneT" refers to "TenneT TSO B.V."

² mFRRda is a specific capacity product for which TenneT has been granted an exemption under Article 26 of the EB-GL. In the future TenneT expects to use two different mFRR products: (1) specific product mFRR, the current product noodvermogen with the implementation of energy bids, (2) standard product mFRR, for which exchanges with other European TSO's takes place via the MARI platform.

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1. Definitions and abbreviations

Term	Description / abbreviation
automatic Frequency Restoration Reserve (aFRR)	aFRR
manual Frequency Restoration Reserve	mFRR
<i>standard</i> manual Frequency Restoration Reserve, directly activated or scheduled activated	mFRR standard product, or also <i>MARI-product</i>
Scheduled Activation (or also Scheduled Activated)	SA
Direct Activation (or also Direct Activated)	DA
Balancing Service Provider	BSP
Balance Responsible Party	BRP
Imbalance settlement period	ISP
Quarter-hour	QH
Current quarter-hour	QH0
Previous quarter hour	QH-1
Beginning of the current QH (or current ISP)	T
Execution day	D
Day prior to the day of execution	D-1
Day after the execution day	D+1
Validity period	The 15-minute period in the bid for which the price / volume combination is applicable.
Cross-border marginal price	CBMP

Table 1 – Definitions and abbreviations

2. Product requirements

The Balancing Service Provider (BSP) for mFRR standard product has the obligation, over the period for which the energy bid is valid, to be able to draw or supply the power agreed from the Dutch grid on call from TenneT, depending on the direction bid in (up- or downward- regulating).

For upward regulation by reducing the electrical load (off-take from the grid) or by increasing production (feed-in to the grid). For downward regulation by increasing the load or by reducing production.

The standard mFRR has the following characteristics and/or conditions:

- Standard mFRR is an active power product
- Standard mFRR energy bids are asymmetric (either up or down)
- Standard mFRR can be provided with a portfolio consisting of one or more technical installations .
- Standard mFRR deliveries are primarily location independent as long as the delivery takes place within TenneT's LFC area (the Netherlands).
- Technical Installations providing standard mFRR must be connected to the national electricity grid.
- A BSP's portfolio has a minimum prequalified capacity of 1 MW
- The minimum size of an energy bid is 1 MW
- The step size of the energy bids is 1 MW
- The full activation time must follow the set delivery profile (see chapter 4)
- The settlement period is 15 minutes
- Standard mFRR includes both DA (direct activated) as well as SA (scheduled activated) (see chapter 4)

Supply balance services in case of restrictions on transport right.

The provision of standard mFRR is seen as a portfolio product. If a connected party is subject to a restriction on his transport right in the context of the emergency and restoration process (for example, a maximum feed-in is in effect) and not meaning restrictions on prognosis as meant in Netcode electricity article 9.2(4), no/not all standard mFRR can be supplied at that location if it exceeds the restriction. The connected party is responsible of the communication of the restriction to the BSP. In case of a restriction on the transport right of connections in the BSP portfolio the BSP should not offer any energy bids (before 25 minutes before ISP start) on connections if this cannot be delivered within the transport right of the connections in its portfolio.

Upward regulating power

Upward regulating power can be provided by reducing the electricity demand from the grid (load reduction) or by increasing the power supply (raising production) into the grid.

For standard mFRR reserved on production units, both the offered power and the ability to ramp up (ramp speed) must be kept free on the portfolio of production units. This means that a production portfolio cannot run at full power.

In addition, the unit must have a stable profile to be able to determine unambiguously both the reference value and the realization of the supplied standard mFRR.

Downward regulating power

Downward regulating power can be provided by increasing the demand from the grid (load) or by decreasing supply (production).

For standard mFRR reserved on production units, both the offered power and the ability to ramp down (ramp speed) must be kept available on the portfolio of production units. This means that a production portfolio may not run at its technical minimum power.

In addition, the unit must have a stable profile to be able to determine unambiguously both the reference value and the realization of the supplied standard mFRR.

Dispatch requirements

TenneT communicates the activation by means of electronic message. In the message the individual energy bid(s) and MW per energy bid to be activated are indicated.

Requirements installations

For the supplied or withdrawn power, it must be explicitly stated through which allocation point(s) the power will be realized. These must be allocation points in the Dutch public electricity grid, not necessarily the TenneT grid.

With a view to verifiability, no standard mFRR can be supplied at an allocation point on which aFRR, incident reserve or any other or energy regulation³ is also active at the same time without separate metering, except for FCR. See the illustration below which configurations are allowed and which are not. If the BSP is delivering standard mFRR with the same unit as with FCR then the burden of proof is on the BSP(s) to demonstrate, if requested, that both calls have been met when required.

³ This includes schemes other than Automatic Frequency Restoration Reserve (aFRR)

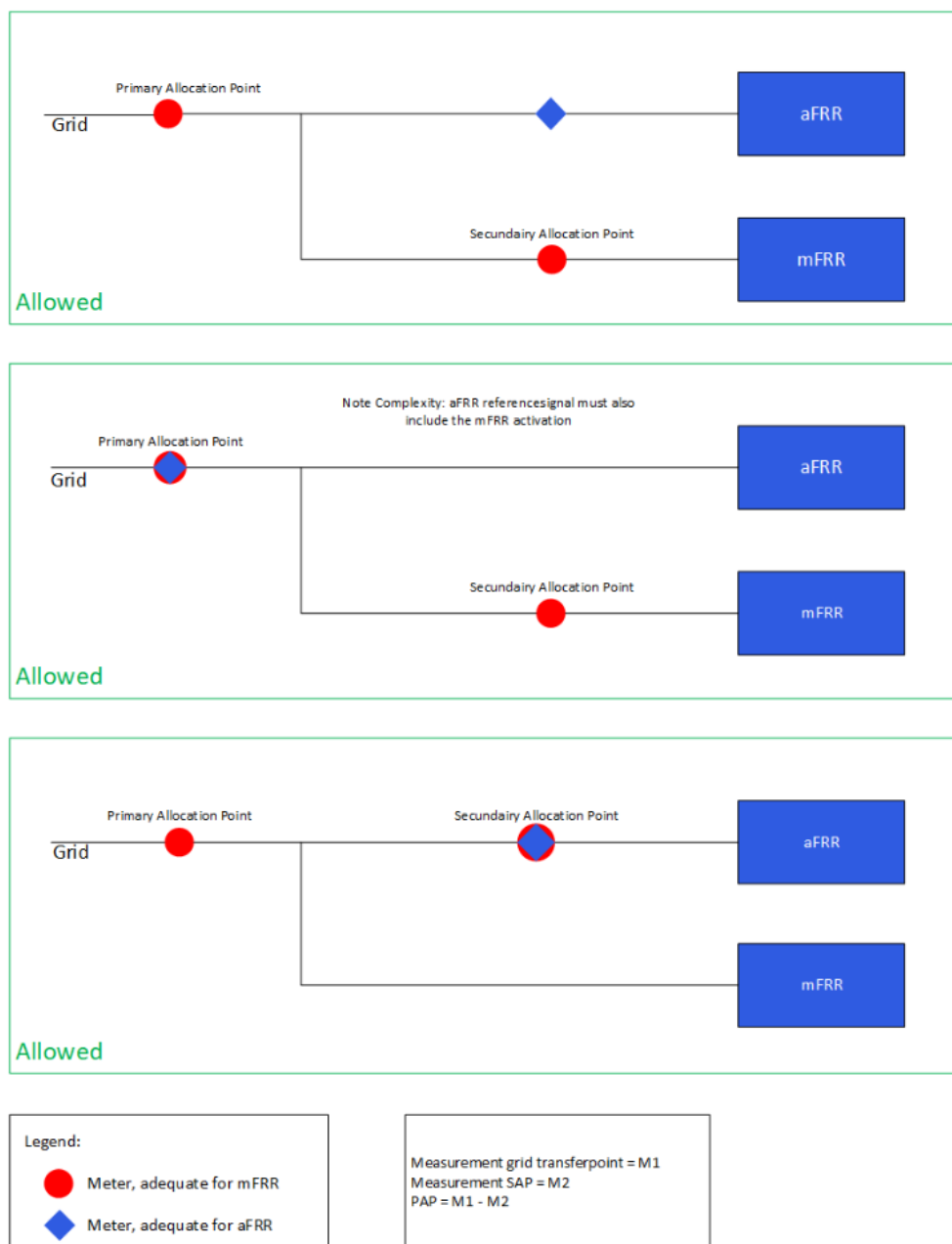


Figure 1 – Allowed configurations for simultaneous delivery of aFRR and mFRR standard product (standard mFRR)

3. Energy bids

The standard mFRR is activated based on energy bids. These energy bids are submitted by a BSP on a voluntary basis. The energy bids which a BSP can send are the simple bids from the bidding options based on the mFRR IF (implementation framework)⁴. The gate closure for sending, changing or withdrawing energy bids is 25 minutes before the start of the ISP for which the bid is valid.

The requirements of standard mFRR energy bids are laid out in the Netcode Bijlage 20 and Bijlage 25.

Energy bids must meet the following characteristics:

- Energy bids have a unique identification per ISP
- Energy bids are valid for 1 ISP (15 minutes)
- Energy bids are asymmetrical (either for upward or downward regulation)
- Energy bids have a bid volume between 1 MW and 9999 MW
- Energy bids can only be fully divisible (see paragraph 3.1)
- Energy bids can be either SA (scheduled activated) or SA & DA (direct activated) (see chapter 4)
- Status of the bid:
 - a. New
 - b. Changed
 - c. Withdrawn

If the availability of assets in the BSP portfolio changes after sending of a balancing energy bid, the BSP must change the already sent balancing energy bid. The change can be the volume and/or price. If the complete volume of the balancing energy bid is no longer available due to the unavailability of the asset(s) the balancing energy bid needs to be withdrawn before gate closure time of the ISP.

Divisibility

According to the standard mFRR bid structure, energy bids can be divisible or indivisible. This can be done by indicating the minimal offered volume.

However, as stated in the ACM approval for the use of incident reserve as a specific product⁵, and also included in the Netcode Bijlage 25⁶:

- the maximum quantity for an indivisible standard mFRR energy bid is 1 MW;
- at the same time, the minimum quantity for a standard mFRR energy bid is 1 MW;
- since these two are equal, the energy bid is in practice fully divisible.

As a result, the only accepted value in the “minimal offered volume” field is 0⁷. If a value different from 0 is received as “minimum offered volume”, the bid will be rejected.

⁴ Implementation framework for the European platform for the exchange of balancing energy from frequency restoration reserves with manual activation in accordance with Article 20 of Commission Regulation (EU) 2017/2195 of 23 November 2017 establishing a guideline on electricity balancing, 30 September 2022.

⁵ <https://www.acm.nl/system/files/documents/besluit-aanvraag-specifieke-producten.pdf>

⁶ <https://zoek.officielebekendmakingen.nl/stcrt-2024-29868.html>

⁷ 0 (and not 1) in the “minimum offered volume” field is linked to the definition of “fully divisible bid” within the MARI platform.

	Fully divisible bid
Divisibility	Divisible
Offered volume	x MW, where $x \geq 1$ MW
Minimum offered volume	z MW, where $z = 0$ MW

Bid linking

Standard mFRR bids offered by BSPs can be either direct activated (DA) or scheduled activated (SA) – see chapter 4 below.

BSPs who submit a bid for QH0 (current QH), do not yet have the knowledge at the gate closure time for QH0 (which is at T-25min, in QH-2), if their bid will be activated in QH-1 (previous QH) in DA. Therefore the linking of bids between QH is needed. This is called technical bid linking.

The figure below depicts the information state for BSPs.

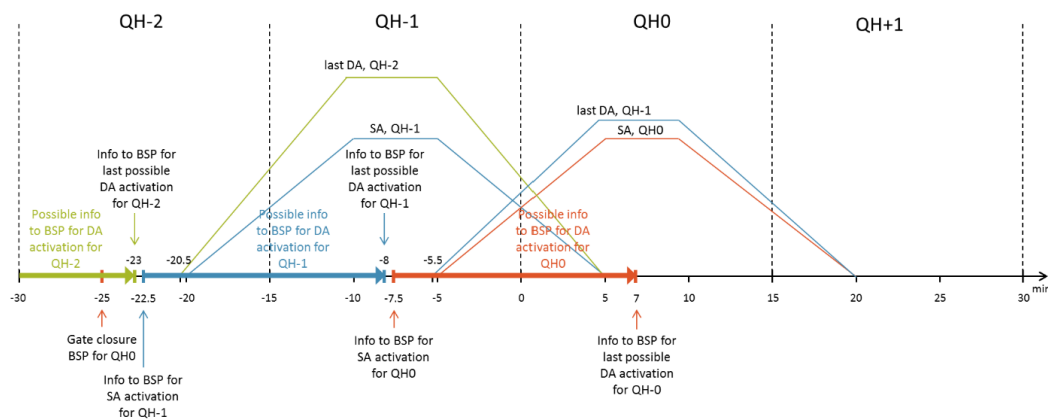


Figure 2 – Information state for BSPs and timelines of possible activations

Technical bid linking is used to prevent unfeasible activations of bids due to direct activation between contiguous ISP's. If a bid is activated in DA in QH-1 and bid linked to a bid in QH0, the SA bid in QH0 will be marked unavailable in the CMOL on the MARI platform (automatically at the platform). This means it cannot be activated for SA in QH0, as it is already delivering the DA activation from QH-1 and therefore technical bid linking prevents infeasible activations.

4. Delivery profile

Because of the allocated cross border capacity the standard mFRR delivery has a strict delivery profile. This means that the delivery of the standard mFRR should be in line with a predefined power profile.

Ramp up and ramp down always last 10 minutes each, for both scheduled activation (SA) and direct activation (DA).

Scheduled activated

Bids to be activated as SA are selected in an AOF (Algorithm Optimization Function) cycle, which is

repeated every 15-minutes within the MARI platform. When a bid is scheduled activated the ramp up always start at T-5 minutes (5 minutes before the start of the ISP the bid is valid for). The delivery period between the end of the ramp up and the start of the ramp down is always 5 minutes. The ramp down period is always 10 minutes. The ramp down period starts 5 minutes before the end of the validity period and ends 5 minutes after the validity period.

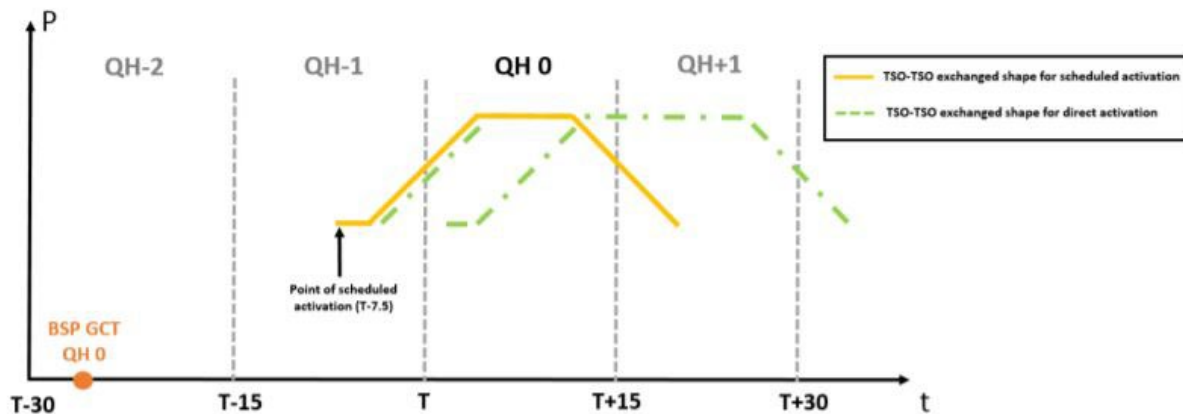


Figure 3 – Requested delivery power profile standard mFRR product

Direct activated

When a DA demand is placed by a TSO at the MARI platform, the AOF selects bids for DA-activation. From the activation message, the start of the ramp up period can be determined. This will be at least 2 minutes after the receipt of the activation message by the BSP. The ramp up period is 10 minutes. Stable power delivery is expected up until 10 minutes into the next ISP. The ramp down period is 10 minutes.

5. Activation process

Activations of standard mFRR energy bids can be sent between T-7.5 minutes and T+7 minutes, where T is the start time of the ISP for which the energy bid is valid. The period in which a bid can be activated depends on the indication in the energy bid. The BSP can indicate that the bid can be scheduled activated (SA), or both scheduled and direct activated (SA & DA).

When there is a need for the standard mFRR⁸ TenneT will receive an activation from the MARI platform. Based on this, the energy bid(s) will be activated with the BSPs.

When sending an activation, the confirmation by the BSP on reception of the activation has to be received by TenneT within 1 minute after sending the activation. If TenneT has not received the confirmation within the specified time period the activation will be considered send. Not responding within the 1 minute period is considered as not complying to the administrative deadlines.

⁸ Resulting from the MARI AOF run, in presence of demand by foreign TSOs and sufficient cross-zonal capacity (CZC) being available.

The information exchange point on BSP side must be available for the period where there is an obligation to deliver: obligation starts 15 minutes before the start of ISPs for which energy bids have been sent to TenneT. Unavailability of the information exchange point on the BSP side will lead to a sanction. The availability of the real time connection is periodically checked.

TenneT will not send a deactivation, the BSP has to deliver in accordance with the specified delivery profile, which includes a deactivation ramp (see chapter 4).

Activated portfolio information

After an activation, the BSP sends an activated portfolio message to TenneT within 5 minutes after receiving the activation. This message contains information on the activated power per allocation point. The activated power is smaller than or equal to the prequalified power on the allocation point(s).

TenneT uses this information to inform the BRP(s) on the allocation point(s) to prevent a counter action of the BRP(s).

6. Data Exchange

To provide standard mFRR , the following messaging must be set up in accordance with the implementation guides. See Figure 2 for the overview of messaging between the various market participants.

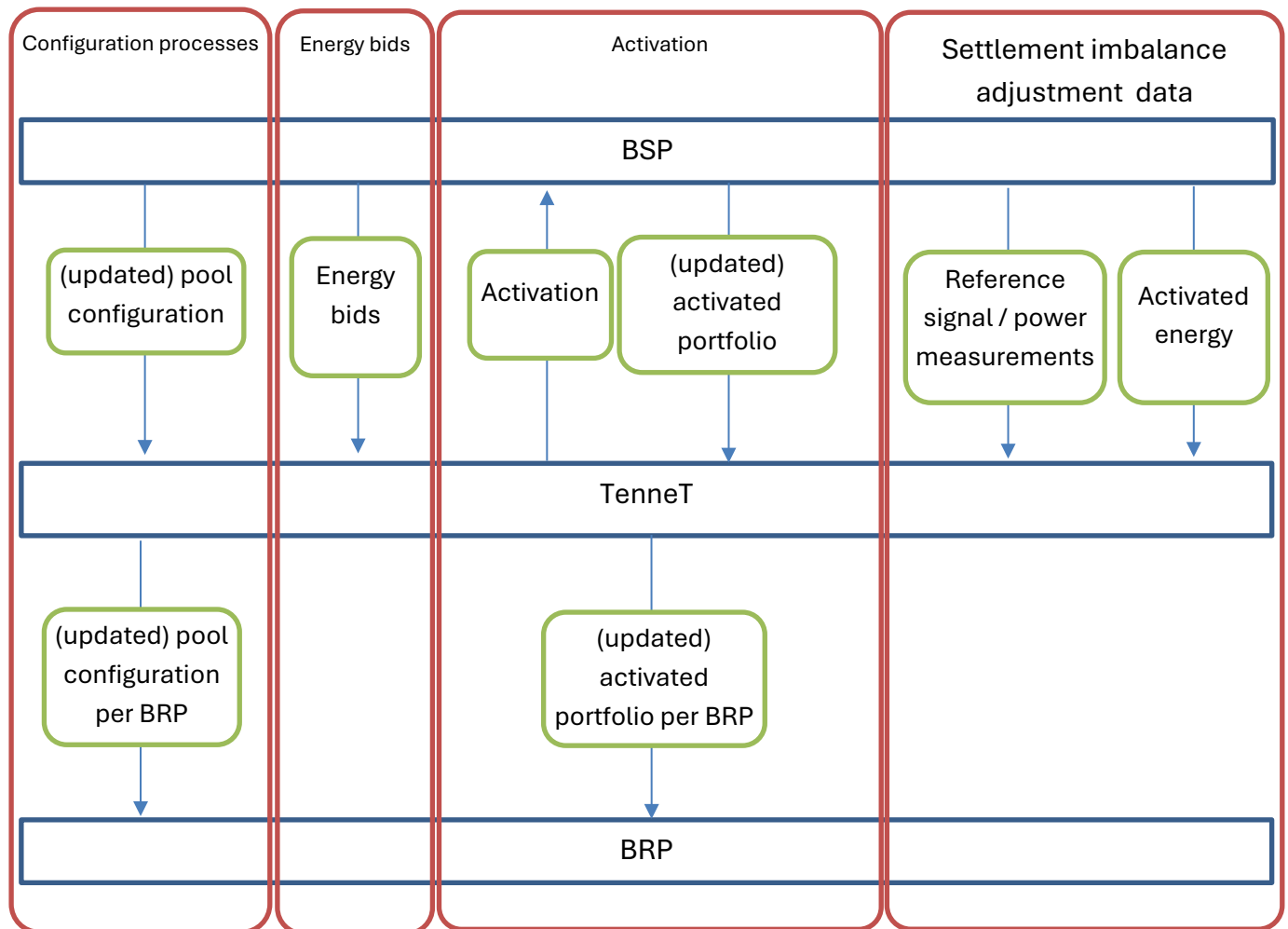


Figure 4 – Overview of message exchanges between market participants

Pool configuration (Configuration process)

The purpose of the pool configuration is to indicate to TenneT which allocation points are active in the BSP's portfolio and which BRPs are active at the various allocation points. TenneT notifies the BRP once a month of all allocation points active in a standard mFRR pool that are attributed to it. The BSP of standard mFRR sends the pool configuration to TenneT in the prescribed manner according to the implementation guide. This message should be sent monthly, no later than 5 working days at 00:00 hour before the start of the month.

In case of interim changes, the BSP sends an update with an effective date at least 2 full days in the future. The last by TenneT accepted message is considered the correct composition of the portfolio of the BSP. A message always includes a complete set of allocation points active in the portfolio of the BSP during the assigned period. Interim changes also include a switch of an EAN-code to another BSP.

The process is as follows:

1. The new BSP sends a request to include a connection in its portfolio. Via filled out PQ Application Form to BSP@tennet.eu.

After internal verification of the connection, the EAN-code receives:

2. The new BSP receives an email from BSP@tennet.eu with confirmation that the EAN-code has been added to his portfolio.
3. The 'old' BSP receives an email (loss message) from BSP@tennet.eu regarding the Allocation Point, the TI which has been switched as of that date and to which BSP.
4. After receiving the loss message, the 'old' BSP is obliged to immediately send an update of the Pool Configuration to TenneT that no longer includes the switched EAN-code.

It is important that the 'old' BSP sends an update of the Pool Configuration to TenneT after receiving the loss message in order to avoid disrupting the rest of the process.

Data exchange on demand (activated portfolio)

The offered power per bid is called by TenneT by means of a message. This message is sent to the BSP, so that it knows how much power to supply. The BSP informs TenneT, within 5 minutes after the request by TenneT, on which allocation points it has activated the called power with what quantity. TenneT immediately informs the BRPs of the allocation points that have been assigned by the BSP with the amount of power that has been activated to avoid counteractions by the BRPs. The information can only contain allocation points which are part of the pool configuration of a BSP. The imbalance adjustment at the BRP of the individual allocation points takes place on the basis of the information in this message (see paragraph 7.1 for the detailed explanation around imbalance adjustment).

Exchanging power data

For the verification of delivery the following data is needed:

- Power measurements of the portfolio (in MW with 4s resolution, + for infeed, - for consumption)
- The reference signal of the portfolio (in MW with 4s resolution, + for infeed, - for consumption)

The power measurements of the portfolio may consist of measurements by the BSP. During prequalification process the BSP proves that the accuracy of the measurements is adequate.

During prequalification process the BSP proves that the reference signal complies with the requirements.

The reference signal and power measurements on portfolio level are sent before 09:00 on D+1. The reference signal and power measurements on portfolio level are sent for the period for which bids are sent, including the ISP prior and two ISPs after the ISP for which the bid is valid.

Activated energy

In the activated energy message the BSP indicates the, by the BSP, requested energy at the allocation points. The sum of the requested energy at the allocation points is equal to the energy TenneT has requested from the BSP.

The following rules apply:

- Only allocation points which are present in the activated portfolio message can be used.
- The requested energy is maximum the calculated energy based on the activated portfolio information, the minimum value is 0 MWh.
- The sum of the activated energy is equal to the requested energy by TenneT.

Example:

If TenneT requests an activation of 30 MW for 15 minutes from a BSP. The BSP divides the requested volume (send in the activated portfolio message) as follows:

- Allocation point A 10 MW
- Allocation point B 20 MW

The calculated energy based on the activated portfolio message:

- Allocation point A 2,5 MWh
- Allocation point B 5 MWh

The activated energy message may contain:

Allocation point	Minimum	Maximum
A	0 MWh	2,5 MWh
B	0 MWh	5 MWh

The requested energy by TenneT is 7,5 MWh (30 MW for 15 minutes). The sum of the activated energy from allocation points A and B must be 7,5 MWh. The energy per allocation point must be between the minimum and maximum values in the table above.

Formats

The BSP shall report the information using the information exchange formats as described in the implementation guides.

Data connection monitoring

Regardless of whether the connection to the BSP is operational, TenneT will activate bids of the BSP when needed. Any lack of response by the BSP will be sanctioned. It is the responsibility of the BSP to monitor the connection themselves.

Data connection testing

When there are changes in the processes or IT of the BSP, depending on the changes these will be tested prior to implementing them in operation. For this, TenneT has an IT environment in which these changes can be tested. If the changes need to be tested with BSPs, these tests with the BSPs can be performed without interfering with operations. This requirement is for all IT connections for the ex-ante, real time, and ex-post.

7. Settlement, imbalance adjustment and penalties

The settlement volumes and imbalance adjustment volumes will be based on the requested energy. The imbalance adjustment will be contributed to the BRPs in the ISPs corresponding to the activated energy information sent by the BSPs.

Determination of settlement and imbalance adjustment volumes

The settlement and imbalance adjustment volumes are determined based on the requested energy.

A fixed profile is used for the delivery of the standard mFRR. See paragraph 4 for more details.

- In case of scheduled activation, the total volume is attributed to the ISP in which the relevant bid was valid
- in case of direct activation, the volumes are attributed to the ISP in which the relevant bid was valid and the subsequent ISP. Any volumes before the validity period are attributed to the validity period. Any volumes after the subsequent ISP are attributed to the subsequent ISP. See Figure 6 below.

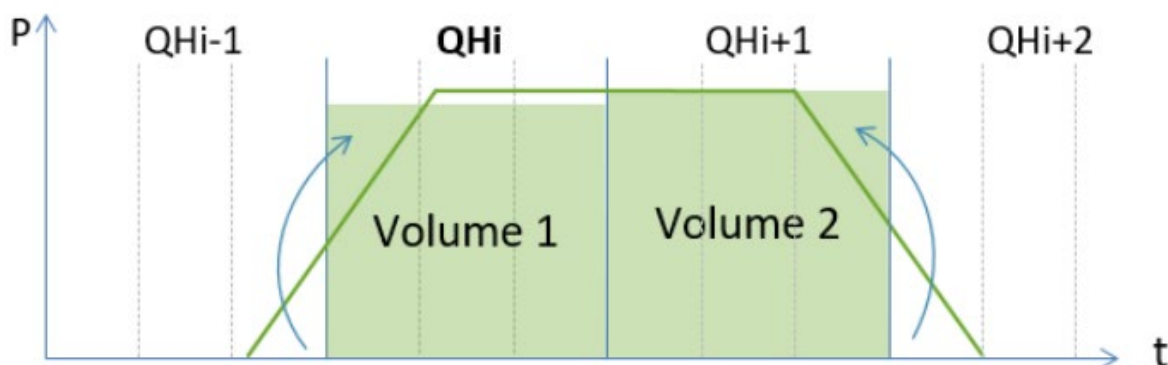


Figure 5 – Determination of settlement and imbalance adjustment volumes

Information exchange between BSP and TenneT for imbalance adjustment

For a standard mFRR activation the BSP needs to deliver the following data to TenneT for the purpose of imbalance adjustment:

Prior to the activation

- *Pool Configuration*: an overview of the allocation points in the BSP pool with corresponding BRP(s);

After the activation

- **Activated Energy:** activated energy by the BSP for standard mFRR per allocation point per 5 minute period.
 - This is the power requested by the BSP to the allocation points, see Figure 5.
 - Only allocation points mentioned in the activated portfolio message can be in the activated energy message
 - The activated energy by the BSP at an allocation point equals the requested energy by TenneT to the BSP

Additionally, the BSP should be able to receive:

- **Confirmation Market Document:** feedback document with information on the settlement volumes and the price used in the settlement.

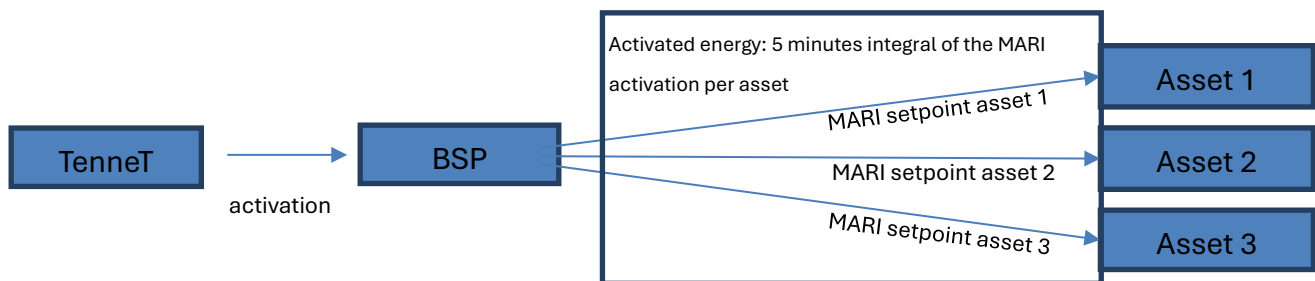


Figure 6 – Illustration activated energy

The energy that the BSP activates (and sends to TenneT in the Activated Energy message) is used for settlement between TenneT and the BSP. The combination of the activated energy message and with the Pool Configuration determines the imbalance adjustment, that TenneT performs for the BRP(s) of the activated allocation points.

The BSP is responsible to divide the activation that it receives from TenneT over the technical installations in its portfolio. The total volume of activated energy should add up to the requested volume by TenneT (based on the activation). If the total of activated energy does not add up to the requested volume the activated energy message will be rejected.

Remuneration

The remuneration for the activation of the standard mFRR depends on the type of activation (SA or DA) and the ISP in which the energy is delivered.

The remuneration of the BSP is calculated as:

$$\text{determined volume (paragraph 7.1)} \times \max(\text{relevant mFRR CBMP, available price of technical linked bid price})$$

With the Confirmation Market Document the BSP is informed on the used price.

Scheduled activated price

In case of SA, the delivered volume is determined for the ISP for which the activation has taken place.

- The volume determined for the ISP is settled at the mFRR CBMP for SA for that ISP.

Direct activated price

In case of DA, delivered volumes are determined for two subsequent ISPs.

- The first ISP (validity period) is settled for the CBMP for DA for that ISP determined by the MARI platform
- The second ISP (the one after the validity period) is settled for either the CBMP for DA for the initial ISP determined by the MARI platform or,
 - For upward regulation: if higher, at the price of a technical linked bid in the second ISP, if available
 - For downward regulation: if lower, at the price of a technical linked bid in the second ISP, if available

Imbalance adjustment

The determined volume per allocation point (paragraph 6.1) is used for the imbalance adjustment with the BRPs.

8. Verification of delivery

As the standard mFRR requires a power profile the verification of delivery is done based on power values/measurements. For this a reference signal with the expected power output and the power measurements are needed, both on portfolio level.

Power data

The general requirements for the power data are:

1. The power data is sent by the BSP to TenneT
2. The unit is MegaWatt [MW], with 3 decimals
3. A new value is generated every 4 seconds
4. Power data (reference signal and power measurements) is sent for the period in which energy bids have been sent, but also including 1 ISP prior to the first ISP for which energy bids have been sent, and 2 ISP after the last ISP for which energy bids have been sent.
5. Sign convention is as follows: + for infeed, - for consumption.

Reference signal

The reference signal is intended to determine the quality of the supplied standard mFRR when an energy bid has been activated by TenneT. The reference signal represents the expected power output of the assets

active in the BSP's portfolio. The reference signal is supplied by the BSP. The reference signal is generated by the BSP in real time and sent to TenneT as described in paragraph 6.3.

For the standard mFRR reference signal the following properties will apply:

1. The reference signal should contain all information how the standard mFRR portfolio would have behaved if standard mFRR bids would not have been activated. This includes information such as other ancillary services and the schedule of assets. The reference signal may be constructed in real-time or ahead of time if that would be preferred by the BSP. The reference signal may not be constructed ex-post.
2. When energy bids have been sent, the reference signal must:
 - a. Represent the power of at least the available installations for delivery
 - b. Represent the power of at most of the complete prequalified portfolio of the BSP. TenneT does want to warn about the "ghost signals" from measuring instruments while the unit is being maintained for example. This can disturb the reference signal and is undesirable.
3. The reference signal should contain the power output of the same installations as the aggregated power measurement.
4. *The reference signal should have the same granularity as (and be time synchronous with) the power measurement.*
 - o *The reference signal should have the same sign convention (+ for infeed, - for consumption) as the power measurement.*
5. The reference signal may not be influenced by the standard mFRR activation sent by TenneT.
6. The reference signal must always be constructed in the same way, independent of the activation. The reason for this is that TenneT wants to be able to check the quality of the reference signal. For this, the reference signal must be compared with the power measurement outside an activation.
7. *The reference signal should be sufficiently accurate since the standard mFRR-delivery will be assumed to be equal to the measurement minus the reference signal. As such, the combination of both should comply with accuracy requirements.*
8. The set-up and used input parameters on how the reference signal is built up are described by the BSP in the prequalification report. Changes in the reference signal from what is described in the prequalification report must be reported.

What is allowed/best practices:

- The predicted power schedule (E-program), including planned ramps between ISP transitions.
- Calculate output of "power controller" by means of input values such as temperature.
- Calculation of predicted power based on reference measurements.
- Full or partial processing of TenneT's activation in the reference signal only after approval by TenneT.

What is not allowed:

- Units that are not prequalified included in the reference signal.
- Mirroring, which means that the reference signal changes in the opposite direction to the activation.

Power measurement portfolio

The BSP must send the power measurement of its portfolio to TenneT with a resolution of 4 seconds.

However, it will not always be possible to perform power measurements per technical installation with such a resolution. It is therefore allowed to set up an aggregated power measurement using measurements per technical installation with a lower resolution and where inter- /extrapolation⁹ takes place. The BSP using this must demonstrate during the prequalification process why the aggregated power measurement is a good representation of the individual measurements.

TenneT must be informed if there is a change in the portfolio that causes a different calculation of the aggregated measurement. An example of this could be another algorithm or aggregation method used for the calculation of the aggregated power. The prequalification report (only the relevant parts) should be adjusted accordingly.

The aggregated measurements must be stored with a resolution of 4 seconds for 6 months. Measurements of individual technical installations must also be stored for 6 months with the resolution at which they are used for the aggregated measurement.

9. Sanctions

No or lack of response

The BSP will receive a penalty in case of lack of response when:

- the IT connection is unavailable
- there is no or a negative response to an activation signal from TenneT.
- does not respond with the full power profile with the acceptable response as described in paragraph 4
- does not deliver the requested energy

Delivering aFRR, incident reserve and the standard mFRR simultaneously at the same allocation point is not allowed. Therefore the delivery of aFRR and incident reserve should be ended before the potential start of the standard mFRR delivery.

Determining lack of response power profile

A lack of response is determined if the power profile of the delivery (i.e. measured power minus reference signal) deviates more than 5% of the requested power (which is the maximum of the requested power profile) in either direction from the delivery profile, throughout the whole ramp-up period, full delivery period, and ramp-down period. See Figure 7 below for a visual example.

Determining lack of response delivered energy

A lack of response is determined if the delivered energy volume is less than 95% of the requested energy.

Timeliness of administrative processes

If not complying to the administrative deadlines for more than 4 times a month an administrative fee will be issued.

⁹ The easiest and most transparent method for this is to keep the measurements of the technical installations constant until a different value is measured. Also known as the zero order hold method.

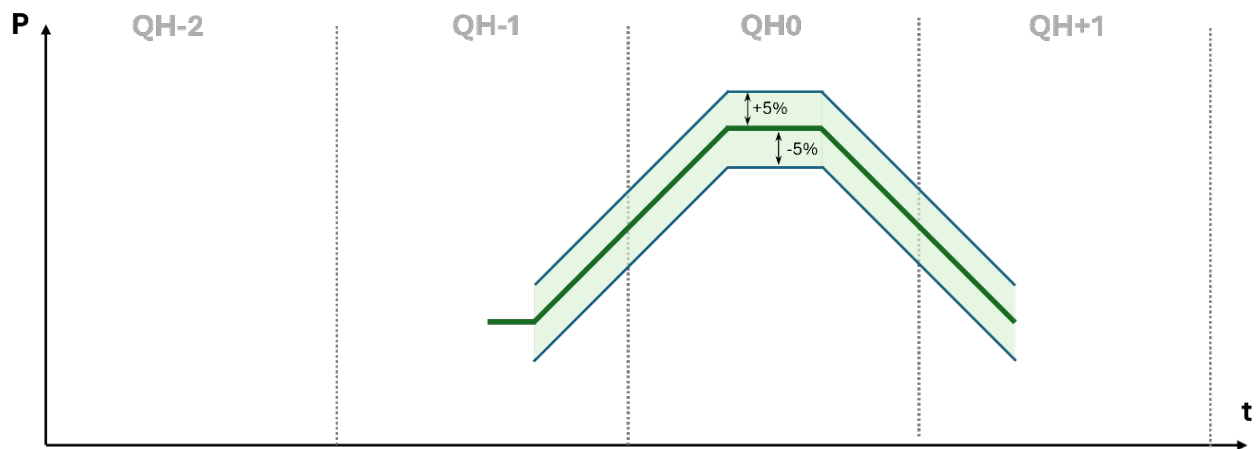


Figure 7 – Example of accepted power corridor in the case of a SA bid

Sanctions

The sanctions will be as follows:

Behavior:	Sanction:
1. Not fulfilling energy delivery obligation	- Reimbursement of the remuneration corresponding to the energy volume not delivered, only in case this was a payment from TenneT to the BSP. This is done considering the highest absolute settlement price of that ISP¹⁰ - Actual sanctioning: 0,1 x Volume not delivered in an ISP x highest absolute settlement price of that ISP¹¹
2. Not following the (power) delivery profile	If 10% of the power measurements are outside the delivery profile of 1 (simultaneous) activation this leads to a sanction of 2.500,00 €
3. Not fulfilling process obligations leads to an administrative sanction of 2.500,00 € a. Manual actions by TenneT in the otherwise automated processes b. Sending ex ante information too late	Administrative sanction of 2.500,00 €
4. Real time information exchange unavailable when there is a delivery obligation	Leads to sanction 1

Table 2 – Applicable sanctions

¹⁰ Reimbursement is always a cash flow from the BSP to TenneT. In case the remuneration of the delivery is a payment from the BSP to TenneT, the BSP will not receive anything back, and the sanctioning will apply on top of that.

¹¹ In case of multiple activations with different prices (for example if a BSP is remunerated the CBMP DA and CBMP SA) the highest price is used for both the reimbursement of the remuneration (when applicable) and the sanctioning. This because in the verification of the delivery the complete delivery of both activations is taken into account. There is no distinction between the SA and DA delivery. The absolute price means that the sanction is always paid to TenneT.

The sanctions above apply on every ISP. There will be only one sanction for the same ISP, meaning that per ISP either sanction 1 will apply or, if higher, a 2.500,00 € sanction.

10. Standard mFRR in a pool

Forming a pool has the advantage for TenneT of potentially offering standard mFRR that otherwise would not be offered. As such, it enlarges the market. A disadvantage, however, is that a pool can complicate administrative and technical procedures for the pool participants, the aggregator (BSP) and TenneT.

The consequences for pool participants are a matter for the aggregator (BSP) and are outside TenneT's area of responsibility.

The table below shows the implications for the parties involved in pooling.

Subject	Pool participant	BSP	TenneT
Pooling	Binding to BSP by mutual agreement. No profile connection		
Technical availability and other technical requirements	Coordinates with BSP so that BSP can ensure required availability	Controls and guarantees compliance with all technical requirements; in particular, availability and full activation time	Handles the same technical requirements as solo providers
Administrative requirements		Arranges and ensures compliance with all administrative requirements; in particular, timely delivery of messages, metering data and coordination with BRPs and pool participants' regular energy suppliers	Handles the same administrative requirements as solo-BSPs
Mandate	Mandating aggregator; this may be arranged in the pool agreement, for example	Starting point is a mandate from all pool participants; should not be a point of discussion in discussions with TenneT	Discussions are being held with BSP
Accessibility		Continuously available	
Call	Calling through BSP	Forwarding of calls by TenneT to pool participants. Inform regular electricity suppliers on connection of pool participants.	Calls to BSP and informs BRPs of connections
Billing / payment	Should be arranged with BSP	Must be arranged with pool participants	Exclusively through aggregator
Balance responsibility	Alignment with BRP. Especially with regard to administrative consequences and possible billing of delivered energy.	Assist pool participant in coordination	The balance responsibility lies with the BRP of the allocation! Communication about the activated power after an activation of the BSP Communicating about implemented imbalance adjustment after a call

Subject	Pool participant	BSP	TenneT
Electricity supplier	Coordinate with electricity supplier. Especially with regard to consequences for supply of energy to the connection and its handling	Assist pool participant in coordination	
Imbalance adjustment	TenneT corrects the imbalance of each pool participant's BRP based on called quantities .		TenneT corrects the called quantity at the connection to the imbalance of the BRP

11. Pre-qualification process

Before a BSP may supply the standard mFRR, the prequalification process must be completed in accordance with the general steps, described in the prequalification procedure manual, published on TenneT's website¹². This chapter further discusses the specific prequalification tests and requirement that must take place for the standard mFRR.

A party is acknowledged as a BSP for energy bids, in parallel to the qualification for their IT connections. This means that there is a single (RPG prequalification) test for the energy product. The BSP is responsible for adhering to the product requirements in this handbook. If, after being acknowledged as BSP for the standard mFRR, the BSP does not meet the product requirements, the acknowledgment can be terminated as described in paragraph 4.2 of the prequalification process published on the TenneT website¹².

Prequalification

In order to prequalify for the supply of standard mFRR, the applicant must carry out a test activation at its own discretion and in coordination with TenneT. The activation is carried out by the applicant himself. Hereby a power to be agreed upon with TenneT, which meets the product requirements from chapter 5. The following applies here:

- The applicant completes and submits an application form (must always be the latest published version) with all Technical Installations (TIs) that it wishes to prequalify (one line per TI) within the applicable deadlines. A correctly filled application form must indicate whether the applicant wants to prequalify for both up- and down-regulating power, or a single direction and the follow-up test(s) must reflect that.
 - In case of application for both upward and downward prequalification, the tests per direction do not necessarily need to be done together, but they must be carried out within the reported test window.

¹² https://tennet-drupal.s3.eu-central-1.amazonaws.com/default/2025-02/20240809_Prekwalificatieproces_NL_0_0.pdf

- Testing outside of an announced (and approved) test window will lead to rejection of the resulting report.
- The tests should at least demonstrate that: the standard mFRR can be delivered in accordance with the delivery profile as specified in chapter 4 of this document
- The power to be activated with the selected TIs during the prequalification test has a power of at least 1MW and must match the setpoint sent out by the applicant. This means that partial activation of selected TIs is allowed during the prequalification test.
- A test activation shows at least compliance with the requirements described in product information standard mFRR. In case the applicant wants to prequalify for up- and down-regulating standard mFRR, a test shall be performed for both directions.

The applicant shall provide the technical and functional specifications of each asset, which, for the purpose of providing standard mFRR, will be used by the applicant.

The applicant must submit a report based on the tests to TenneT describing the test(s) and their results. Both the report and corresponding data must have the structure, prescribed by TenneT, and describe at least the following:

- Date and time of the test.
- EAN-codes and/or names of the assets used as mentioned in the application form.
- Time-synchronous values of the transmitted setpoint, the measured power with a sampling rate of no more than 4 seconds as raw data and in graph form;
- Analyses in which, based on the data, it is illustrated that:
 - The delivery of power is within the power delivery profile (see chapter 4)
 - The response is in line with the activated volume of energy

The BSP must store the original measurement data (including underlying TI data) of the test and keep it for 5 years¹³, or until re-prequalification. TenneT can request this data for verification and control purposes.

12. (Operational) Communication

TenneT will need to communicate with all qualified BSPs for several reasons, i.e.:

- Operational issues
- Changes in the handbook
- Invitations to workshops on current and future developments

For this communication TenneT will use the contact information registered in MyTenneT. It is therefore expected that the BSP keeps the contact information in MyTenneT up to date at all times. A guide on how to update contact information can be found in the FAQ of MyTenneT → my organization.

¹³ 5 years since the prequalification is valid for 5 years.

13. Reference and Resource List

See below for other relevant documents and data sources related to standard mFRR delivery.

General information about support services:

[Support services \(Netherlands\) \(tennet.eu\)](https://www.tennet.eu/support-services-netherlands)

Grid code electricity

The codes are designed to give substance to existing laws and regulations and describe the way network operators behave among themselves and toward other connected parties:

<https://www.acm.nl/nl/onderwerpen/energie/codes-energie/overzicht-codes-energie/> or through

<https://wetten.overheid.nl> .

Imbalance pricing system

Details of the roles, responsibilities and methods applied in relation to the imbalance pricing system.

This document is available at <https://www.tennet.eu/nl/de-elektriciteitsmarkt/ondersteunende-diensten-nederland/documenten-algemeen>.

Implementation Guides

Many IT-related documents concerning the various communication channels with TenneT can be found on MyTenneT. This is a web portal for BSPs which is made accessible after requesting prequalification.

Balance-Delta publication:

<https://www.tennet.eu/nl/de-elektriciteitsmarkt/transparantie-nederland/balans-delta>